Supplementary material for: A finite-size crossover criterion for shadow-based moment estimation, with a hardware case study

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Abstract

This document collects the material the main text refers to but does not print: the hardware protocol, the weight-resolved readout crosstalk measurements, the two session tables, the construction of the single-copy anchor, the cloud-platform credit accounting, and the full listing of resolved crossover cells. Every section and appendix reference below is to the main text.

S1. HARDWARE PROTOCOL

This section reproduces the hardware protocol referred to by the main text.

Platform. Open Quantum (Quantum Rings Inc.) [1], Public Tier, Standard queue, the backend reported by the platform as Rigetti Cepheus-1-108Q. Jobs were submitted as OpenQASM 3 with physical-qubit addressing (`$0`, `$1`, and so on), which the platform required because it rejects non-contiguous virtual registers. The Public Tier routes jobs through the Quantum Compute subnet (SN48) on the Bittensor network, operated by qBitTensor Labs, with validator spot-checking of execution on the target QPU [1]; see Section 6.1 for the interpretive consequences.

Registers. GHZ ladders, verified by re-transpilation to require zero routing SWAPs:

n	physical qubits	CZ gates
2	$\{0,1,9,10\}$	4
3	$\{0,1,2,9,10,11\}$	7
4	$\{0,1,2,3,9,10,11,12\}$	10

The ladders nest, so a single readout characterization on the $n=4$ register covers all three.

Discipline. Every prediction was computed, printed, and committed to version control before the corresponding measurement job was submitted. Raw counts were committed verbatim before any analysis. Where a prediction failed, the model was not adjusted to fit. Every hardware job was quote-gated against a hard credit ceiling before charging.

Total hardware expenditure. Approximately 115 platform credits across nine experimental campaigns.

License conditions. Second, use of the Public Tier carries two license conditions:

publications resulting from work on the tier are required to cite the platform paper [1], which we do, and circuits, results, and metadata are contributed in anonymized aggregated form to a common repository maintained by Open Quantum [1].

S2. READOUT CROSSTALK, WEIGHT-RESOLVED

The main text reports that the measurement crosstalk on this device saturates with excitation weight rather than accumulating. The weight-resolved series behind that statement, and its consequence for the larger registers, are as follows.

The correlation saturates rather than accumulating. Measured across excitation weights $w = 0, 2, 4, 6$, qubit q_0's $P(1|0)$ runs 0.020, 0.169, 0.242, 0.224: a steep rise from $w = 0$ to $w = 2$, then a plateau. A linear extrapolation from the first two points predicts 0.473 at $w = 6$, overestimating the measured value by 0.25. Because the correlation saturates, the crosstalk must be measured at each excitation weight rather than extrapolated; extrapolating instead would have made the $n = 4$ predictions pessimistic by five points.

S3. THE BRACKETED SAME-SESSION EXPERIMENT

Measured collective purities against the prediction bands committed in advance from the opening and the closing readout calibration of the same session.

n	measured	opening band	closing band
2	0.686 [0.672, 0.700]	0.718 – 0.742	0.704 – 0.728
3	0.378 [0.360, 0.397]	0.597 – 0.634	0.593 – 0.631
4	0.420 [0.402, 0.439]	0.506 – 0.553	0.520 – 0.568

All three cells fall below both bands rather than between them, so drift within the session does not account for the gap. The $n = 2$ cell here (0.686, 10,000 shots) is the session-study SWAP test, a different circuit and run from the 0.7184 Bell-purity anchor above, so the two are not repeated readings of one quantity. The degradation is also non-monotonic: $n = 3$ is worse than $n = 4$, despite the $n = 4$ register containing the $n = 3$ register. Neither readout error, which scales with the $2n$ measured qubits, nor CZ count, which grows with n, can produce that ordering.

S4. THE THREE-SESSION COMPARISON

The cross-session picture explains it. Running the same byte-identical circuits, on the same physical qubits, in different sessions:

register	session 1	session 2	session 3
$n = 3$	0.317	0.378	0.587
$n = 4$	0.431	0.420	0.382

S5. THE SINGLE-COPY ANCHOR

The construction of the fifteen-basis anchor, why its statistic is not a purity, and why second-moment agreement between the Pauli and the Haar ensemble does not repair its within-circuit dependence.

The single-copy route failed the same way. Alongside the collective measurements we ran a fifteen-basis single-copy anchor at $n = 2$, whose U-statistic our shadow pipeline predicted at 1.500 ± 0.028 from the same device parameters. That statistic is not a purity: the fifteen bases collapse to eight distinct unitaries, so about 15% of the snapshot pairs share one, which biases the estimator upward, here past the physical bound $\mathrm{Tr}(\rho^2) \leq 1$. The settings are random Pauli-basis measurements, of which there are nine at $n = 2$, so repeats among fifteen draws are an expected consequence of sampling a finite set rather than a flaw, and the 1.500 prediction is locked for this Pauli-basis pipeline. Because the same estimator is applied to the simulated and the measured data, the anchor tests the pipeline rather than measuring a purity. The anchor samples random Pauli bases while the simulation draws local Haar shadows, but the two ensembles share a second-moment matrix and hence carry the same predicted statistic (Section 3.5); that second-order agreement does not repair the anchor's within-circuit repetition, which induces the dependence an i.i.d. U-statistic assumes away, so the anchor tests this pipeline rather than the standard estimator. The device returned 1.344, with a 95% bootstrap interval of [1.266, 1.433] over 9000 snapshots. The gap is 5.6 times the spread the model predicts for a single run of this anchor, and 3.7 times the wider spread the data themselves show; the intervals are disjoint on either accounting — both spreads are conditional shot noise, hence lower bounds on the total uncertainty (Appendix C) — and no simulated run of the locked model reaches the measured value. We do not adjust the model to fit. Three signals therefore fail together in this session: the collective purities at $n = 3, 4$ fall below their bands, their ordering is non-monotonic, and the single-copy pipeline does not reproduce its own anchor. One common cause accounts for all three, and it is the one this section has been describing: the device parameters were characterized in earlier sessions, and the effective execution channel changed between sessions — byte-identical QASM does not guarantee an identical pulse schedule, and the decentralized route cannot be verified per job. That two protocols with different circuits and different estimators both miss their locked predictions in the same session is what makes drift, rather than any protocol-specific defect, the parsimonious explanation.

S6. CLOUD-QPU ECONOMICS

One further constraint emerged that affects reproducibility rather than physics. The single-copy shadow route requires many independent random measurement bases. On a per-circuit-priced cloud platform without circuit batching, each basis is a separately billable job. An unbiased shadow purity estimate at the precision needed for this comparison requires on the order of a few hundred independent random measurement bases, each a separately billable job – one to two orders of magnitude more than the roughly 3 credits per cell the collective route used at the same shot count.

Single-copy classical shadows are therefore economically infeasible under the per-circuit pricing and batching model of the cloud platform we used, at the scale required to reproduce their own sample-complexity claims. This is the reason our hardware single-copy baseline at $n \geq 3$ is computed from independently measured device parameters rather than measured directly, with a fifteen-basis $n = 2$ anchor as the one point we could afford to measure. That anchor did not reproduce its prediction (Section 6.4), so the computed points are model output that the hardware did not corroborate, and we present them as such.

S7. THE FULL CROSSOVER TABLE

TABLE I: All 83 resolved crossover cells. “mult” is the copy budget as a multiple of the 2000 baseline. A cell counts as within one qubit (✓) when $|\,n^*_{\mathrm{pred}} - n^*_{\mathrm{meas}}\,| \leq 1$.

ensemble	k	channel	rate	mult	n^*_{meas}	n^*_{pred}	≤ 1
noisy-pure	2	amp. damp.	0.00	1	2	2	✓
noisy-pure	2	amp. damp.	0.02	1	2	2	✓
noisy-pure	2	amp. damp.	0.05	1	6	6	✓
noisy-pure	2	amp. damp.	0.05	1	6	6	✓
noisy-pure	2	amp. damp.	0.05	4	8	8	✓
noisy-pure	2	amp. damp.	0.10	1	7	7	✓
noisy-pure	2	amp. damp.	0.10	1	7	7	✓
noisy-pure	2	amp. damp.	0.10	4	9	9	✓
noisy-pure	2	dephas.	0.00	1	2	2	✓
noisy-pure	2	dephas.	0.02	1	2	2	✓
noisy-pure	2	dephas.	0.05	1	6	6	✓
noisy-pure	2	dephas.	0.05	1	6	6	✓
noisy-pure	2	dephas.	0.05	4	8	8	✓
noisy-pure	2	dephas.	0.10	1	7	7	✓
noisy-pure	2	dephas.	0.10	1	7	7	✓
noisy-pure	2	dephas.	0.10	4	9	9	✓
noisy-pure	2	depol.	0.00	1	2	2	✓
noisy-pure	2	depol.	0.02	1	2	2	✓
noisy-pure	2	depol.	0.05	1	2	2	✓
noisy-pure	2	depol.	0.05	1	2	2	✓
noisy-pure	2	depol.	0.05	4	5	5	✓
noisy-pure	2	depol.	0.05	16	7	7	✓
noisy-pure	2	depol.	0.10	1	5	5	✓
noisy-pure	2	depol.	0.10	1	5	5	✓

ensemble	k	channel	rate	mult	n^*_{meas}	n^*_{pred}	≤ 1
noisy-pure	2	depol.	0.10	4	7	7	✓
noisy-pure	2	depol.	0.10	16	8	8	✓
noisy-pure	3	amp. damp.	0.00	1	2	2	✓
noisy-pure	3	amp. damp.	0.02	1	2	2	✓
noisy-pure	3	amp. damp.	0.05	1	8	7	✓
noisy-pure	3	amp. damp.	0.05	1	8	7	✓
noisy-pure	3	amp. damp.	0.10	1	8	8	✓
noisy-pure	3	amp. damp.	0.10	1	8	8	✓
noisy-pure	3	dephas.	0.00	1	2	2	✓
noisy-pure	3	dephas.	0.02	1	2	2	✓
noisy-pure	3	dephas.	0.05	1	8	7	✓
noisy-pure	3	dephas.	0.05	1	8	7	✓
noisy-pure	3	dephas.	0.10	1	8	8	✓
noisy-pure	3	dephas.	0.10	1	8	8	✓
noisy-pure	3	depol.	0.00	1	2	2	✓
noisy-pure	3	depol.	0.02	1	2	2	✓
noisy-pure	3	depol.	0.05	1	2	2	✓
noisy-pure	3	depol.	0.05	1	2	2	✓
noisy-pure	3	depol.	0.05	4	4	4	✓
noisy-pure	3	depol.	0.05	16	8	8	✓
noisy-pure	3	depol.	0.10	1	4	4	✓
noisy-pure	3	depol.	0.10	1	4	4	✓
noisy-pure	3	depol.	0.10	4	8	7	✓
noisy-pure	4	amp. damp.	0.00	1	2	2	✓
noisy-pure	4	amp. damp.	0.02	1	2	2	✓
noisy-pure	4	amp. damp.	0.05	0.25	2	4	×
noisy-pure	4	amp. damp.	0.05	1	8	8	✓
noisy-pure	4	amp. damp.	0.10	0.25	6	6	✓
noisy-pure	4	amp. damp.	0.10	1	8	8	✓

ensemble	k	channel	rate	mult	n^*_{meas}	n^*_{pred}	≤ 1
noisy-pure	4	dephas.	0.00	1	2	2	✓
noisy-pure	4	dephas.	0.02	1	2	2	✓
noisy-pure	4	dephas.	0.05	0.25	2	2	✓
noisy-pure	4	dephas.	0.05	1	8	8	✓
noisy-pure	4	dephas.	0.10	0.25	6	6	✓
noisy-pure	4	dephas.	0.10	1	8	8	✓
noisy-pure	4	depol.	0.00	1	2	2	✓
noisy-pure	4	depol.	0.02	1	2	2	✓
noisy-pure	4	depol.	0.05	0.25	2	2	✓
noisy-pure	4	depol.	0.05	1	2	2	✓
noisy-pure	4	depol.	0.05	1	2	2	✓
noisy-pure	4	depol.	0.05	4	3	2	✓
noisy-pure	4	depol.	0.10	0.25	2	2	✓
noisy-pure	4	depol.	0.10	1	3	2	✓
noisy-pure	4	depol.	0.10	1	3	2	✓
haar-pure	2	depol.	0.05	1	2	2	✓
haar-pure	2	depol.	0.10	1	5	5	✓
low-rank	2	amp. damp.	0.05	1	6	6	✓
low-rank	2	amp. damp.	0.10	1	6	6	✓
low-rank	2	dephas.	0.05	1	6	6	✓
low-rank	2	dephas.	0.10	1	6	6	✓
low-rank	2	depol.	0.05	1	2	2	✓
low-rank	2	depol.	0.10	1	4	4	✓
low-rank	2	depol.	0.30	1	6	6	✓
ghz-noisy	2	amp. damp.	0.05	1	6	6	✓
ghz-noisy	2	dephas.	0.05	1	6	6	✓
ghz-noisy	2	dephas.	0.10	1	6	6	✓
ghz-noisy	2	depol.	0.05	1	2	2	✓
ghz-noisy	2	depol.	0.10	1	5	4	✓

ensemble	k	channel	rate	mult	n^*_{meas}	n^*_{pred}	≤ 1
ghz-noisy	2	depol.	0.30	1	6	6	✓

[1] Bob Wold, Omar Armbruster, and Ryan Kuhn, *Open Quantum: Democratizing Access to Quantum Computing Resources*, white paper, Quantum Rings Inc., Broomfield, CO (2026), https://www.openquantum.com/whitepaper.pdf (accessed July 2026; archived at https://web.archive.org/web/20260212195549/https://www.openquantum.com/whitepaper.pdf).